Shravan Chandra

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Summary

ML Software Engineer with 5+ years formulating rigorous analytical approaches to ambiguous engineering problems and shipping them as production-grade ML and agentic GenAI systems at scale, including 80+ TB telemetry pipelines and real-time anomaly detection at Bosch. Experienced in Python and SQL across the ML lifecycle, from distributed Databricks/PySpark infrastructure to agentic system design, vector-based retrieval, and optimized LLM inference. Maintains an active, IEEE-published research thread in sign language translation, specializing in VQ-VAE tokenization and Conformer architectures for accessibility-focused AI.

Experience

Senior ML Engineer, Bosch Global Software Technologies — Bangalore, IN — Jan 2025 – June 2026

- Orchestrated Valfred, a multi-agent GenAI data assistant for natural-language querying of Bosch's testing platforms; parallel SQL agents and response synthesis with vector-based schema retrieval and self-healing execution cut routine data retrieval from 10–15 minutes of manual SQL to a single query, reaching ∼81% eventual success (65% first-pass), deployed via MLflow serving and Unity Catalog on Databricks.
- Architected a distributed Databricks batch pipeline ingesting 80+ TB of high-frequency ECU sensor telemetry; cut per-partition processing time by 50%+ via PySpark query rewrites and broadcast join optimization.
- Developed a real-time Autoencoder-based anomaly detection system on streaming sensor data; reduced component validation cycles by 4 months and delivered €300K+ in annual savings by catching failures before test completion.
- Drove ML architecture decisions and deployment workflows across India, Germany, and China as technical SPOC for 5+ concurrent initiatives. Mentored 3 interns to full-time ML roles; authored onboarding guides and infrastructure standards adopted team-wide.

ML Engineer, Bosch Global Software Technologies — Bangalore, IN — Aug 2023 – Dec 2024

- Built a real-time predictive failure pipeline using Elastic Net regression on AUTOSAR DEM debounce signals; achieved 0.92 recall and surfaced early-warning alerts 30 seconds before ECU system errors.
- Implemented online distribution-shift detection (KL divergence + hypothesis testing) across 100+ ECUs, flagging sensor anomalies with 95% accuracy under continuous high-frequency data ingestion.
- Led TensorFlow 1.x → PyTorch migration across a multi-GPU Azure environment, achieving 48% faster training throughput and unblocking distributed training at scale.

Software Engineer, Bosch Global Software Technologies — Bangalore, IN — Aug 2021 – Aug 2023

- Built sensor analytics platform with real-time visualization and time-series forecasting; automated data pipelines reducing manual processing time by 45% across 25+ engineers.

Education

University of British Columbia — Master of Data Science, Vancouver, Canada — Aug 2026 – Jun 2027

PES University — BTech in Electrical & Electronics Engineering | CGPA: 8.5/10.0 — Aug 2017 – May 2021

Projects

ASL Guide — Production Accessibility App, 2025–2026 — github/ASL-App

- Designed a dual-agent agentic system (LangGraph) for real-time text-to-ASL translation, automating 10+ ASL grammar rules with structured glosses for handshape, motion, and non-manual markers.
- Reduced LLM token costs by 73% via Gemini Context Caching; built dual-layer caching (Redis L1 + in-memory LRU L2) to serve low-latency responses under concurrent load without cold-start penalties.
- Engineered a TensorFlow.js fingerspelling classifier (95.5% accuracy, 36 classes) running entirely in-browser via MediaPipe, achieving sub-100ms inference latency with no server round-trips.
- Deployed with Docker, PostgreSQL, and async SQLAlchemy; implemented rate limiting, security headers, and an admin dashboard for usage analytics and feedback triage.
- *Stack: FastAPI • TensorFlow.js • LangGraph • Gemini API • Redis • PostgreSQL • Docker • MediaPipe • ONNX*

Research & Publications

A sustained research program in sign language recognition — each stage informing the next toward a generalizable ASL translation system. github/Sign-Language-Translation

Stage 3: Neural Sign Language Translator (VQ-VAE + Conformer Seq2Seq) 2026 — In Progress

- Designing a Factorized VQ-VAE discretizing 3D landmark sequences into four hierarchical factors (Pose, Motion, Dynamics, Facial Expressions); built a Conformer Seq2Seq with Hybrid CTC + Attention decoder and Cross-Factor Attention for NMM-hand coordination.
- Engineered a signer-independent pipeline with Hand Dominance Normalization and a fallback-based coordinate preprocessor to improve generalization across unseen signers and camera setups.

Stage 2: Graph-Based Recognition via ST-GCN 2025

- Experimented with ST-GCN modeling 209 skeletal joints with Edge Importance Weighting, and Spatial Attention; graph-based spatial modeling proved bottlenecked by signer diversity on the Google ASL Signs dataset — a finding that directly motivated the shift to learned discrete tokenization in Stage 3.

Stage 1: Landmark Vision Transformer — Google ASL Signs (250 classes) 2024

- Achieved **80.85% top-1 accuracy** on Google ASL Signs (250 classes, 94K samples) using a Landmark Vision Transformer on 543-keypoint landmark data; engineered body-relative velocity features, signer-independent splits, Mixup + Finger Dropout augmentation, and 5-pass TTA ensemble inference.

Dynamic Sign Language Translator — IEEE ICCAR 2022 May 2022

Shravan Chandra, Venkatarangan MJ, Jyothi TN 10.1109/ICCAR55106.2022.9782661

- Published lightweight sign recognition system (MediaPipe + XGBoost) achieving 90%+ accuracy across 15 ASL phrases at IEEE International Conference on Control, Automation and Robotics.

Skills

Languages: Python, JavaScript, TypeScript, SQL, R

ML / DL: PyTorch, TensorFlow, TensorFlow.js, Scikit-Learn, Transformers, OpenCV, MediaPipe, ONNX

GenAI: LangChain, LangGraph, RAG Systems, Prompt Engineering, AI Agents, LLM APIs, Vector DBs (Chroma)

Data & Cloud: PySpark, Databricks, Azure ML, Pandas, Power BI, MLflow

Infra / Tools: Docker, FastAPI, PostgreSQL, Redis, Git, Kubernetes, Jupyter